

Practice Questions for 15L2-SAT

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1 Questions

1. According to the lecture, what is the main focus of progress in SAT solving over the last 20 years?
 - A. Developing faster hardware
 - B. Increasing the number of variables and constraints that can be handled
 - C. Improving the user interface of SAT solvers
 - D. Reducing the cost of SAT solvers
 - E. Making SAT solvers more accessible to non-experts
2. In the context of SAT solving, what does a "partial assignment" refer to?
 - A. An assignment where all variables are assigned a truth value
 - B. An assignment where no variables are assigned a truth value
 - C. An assignment where some variables are assigned a truth value, and others are not
 - D. An assignment that only considers a subset of the clauses
 - E. An assignment that is only valid for a specific part of the search space
3. According to the lecture, what is the purpose of the "Pure Literal Rule" in SAT solving?
 - A. To identify clauses that are always true
 - B. To identify clauses that are always false
 - C. To assign truth values to variables that only appear positively or negatively in the formula
 - D. To simplify the formula by removing redundant clauses
 - E. To identify unit clauses

2 Answers

1. According to the lecture, what is the main focus of progress in SAT solving over the last 20 years?
 - A. Answer A is incorrect because the lecture states that while Moore's Law helped, it was not the main reason for the progress.
 - B. Answer B is correct because the lecture specifically mentions the increase in problem size from 100 variables and 200 constraints to 1,000,000+ variables and 5,000,000+ constraints.
 - C. Answer C is incorrect because the lecture does not discuss the user interface of SAT solvers.
 - D. Answer D is incorrect because the lecture does not discuss the cost of SAT solvers.
 - E. Answer E is incorrect because the lecture does not discuss the accessibility of SAT solvers to non-experts.
2. In the context of SAT solving, what does a "partial assignment" refer to?
 - A. Answer A is incorrect because a partial assignment does not assign all variables a truth value.
 - B. Answer B is incorrect because a partial assignment assigns truth values to some variables.
 - C. Answer C is correct because a partial assignment is defined as a function that assigns truth values to some variables, leaving others undefined.
 - D. Answer D is incorrect because a partial assignment considers all clauses, but may not have truth values for all variables.
 - E. Answer E is incorrect because a partial assignment is not specific to a part of the search space.
3. According to the lecture, what is the purpose of the "Pure Literal Rule" in SAT solving?
 - A. Answer A is incorrect because the Pure Literal Rule does not identify clauses that are always true.
 - B. Answer B is incorrect because the Pure Literal Rule does not identify clauses that are always false.
 - C. Answer C is correct because the Pure Literal Rule assigns a truth value to variables that only appear positively or negatively in the formula.
 - D. Answer D is incorrect because the Pure Literal Rule does not remove redundant clauses.
 - E. Answer E is incorrect because the Pure Literal Rule does not identify unit clauses.